General Mathematics — Exercise Sheet 20

Inversion and determinant definitions

Variant: Statements

Exercise 1 (A parameter in two possible pivots). For $t \in \mathbb{R}$, consider

$$A_t = \begin{pmatrix} 1 & 1 & 0 \\ 0 & t & 1 \\ 0 & 0 & t-1 \end{pmatrix}.$$

Find every value of t for which A_t is invertible, using Lecture 20's elimination criterion. Construct its inverse by Gauss–Jordan operations on $[A_t \mid I_3]$ whenever it exists. For each exceptional parameter, exhibit a nonzero homogeneous solution as the obstruction. Do not use a determinant criterion from a later lecture.

Exercise 2 (A matrix unknown on each side). Let

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}.$$

Construct A^{-1} by elimination. Find every $X \in \mathbb{R}^{2\times 2}$ satisfying $AX = B$ and every $Y \in \mathbb{R}^{2\times 2}$ satisfying $YA = B$. Explain why the two answers need not be equal, and verify both equations. Use the matrix product rules of Lecture 19.

Exercise 3 (A one-sided inverse). Let $A, C \in K^{n\times n}$, with $K = \mathbb{R}$ or $\mathbb{C}$ and $n \geq 1$. Suppose $AC = I_n$. Prove $CA = I_n$ using the homogeneous-system criterion of Lecture 20, without determinants. Then test the rectangular pair

$$U = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Compute UV and VU and explain why the square hypothesis matters.

Exercise 4 (The defining column, including complex entries). Compute the determinants of

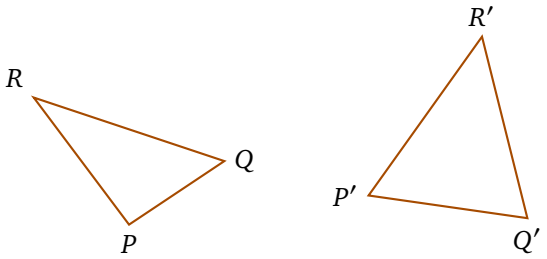
$$D = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 3 \\ -1 & 0 & 2 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 1+\mathrm{i} & 2 & 0 \\ 0 & 1-\mathrm{i} & 4 \\ 0 & 0 & \mathrm{i} \end{pmatrix}.$$

For D, use recursive first-column expansion and display the two nonzero 3×3 minors. For H, use the upper-triangular formula proved from that definition. Do not use row-operation determinant rules from Lecture 21.

Exercise 5 (Area after an affine change). In the real coordinate plane, let $P = (1, -1)$, $Q = (4, 1)$ and $R = (-2, 3)$. Let the map $T : \mathbb{R}^2 \to \mathbb{R}^2$ be given by

$$T(x) = Mx + p, \qquad M = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}, \quad p = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Compute $P' = T(P)$, $Q' = T(Q)$ and $R' = T(R)$. Use the signed-area determinant from Lecture 20 to find the areas and orientations of both ordered triangles. Use the elementary fact that a triangle has half the area of the parallelogram on its two side columns. Explain directly why the translation column p cancels from those columns. Compute the ratio of the two areas for these data without using a general multiplicativity theorem for determinants.



The drawings show the original and transformed triangles on separate scales.